Long exact sequence of homotopy groups of a pointed map

The Univalent Foundations Program

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Editorial note. Only the selected result is counted. Its local constructions and author proof are retained without generated proof completion.

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Allowed standard prerequisites (editorial).

- Univalent homotopy theory foundations. Univalence, function extensionality, dependent path transport, higher inductive pushouts/suspensions, truncations and basic pointed homotopy notions. The selected local code, fiber or gluing constructions are retained as author proof. Original numbered references identify these foundations.

Source reference. Original exercise 8.5: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1412–1412. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original exercise 2.10: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1402–1402. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original exercise 8.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1383–1383. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original exercise 8.6: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1342–1342. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original definition 7.6.3: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1172–1172. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original theorem 7.3.12: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1856–1856. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original theorem 2.1.6: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1767–1767. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

0.1 Fiber sequences and the long exact sequence

If the codomain of a function $f : X \to Y$ is equipped with a basepoint $y_0 : Y$, then we refer to the fiber $F :\equiv \mathsf{fib}_f(y_0)$ of f over y_0 as **the fiber of** f. (If Y is connected, then F is determined up to mere equivalence; see original exercise 8.5.) We now show that if X is also pointed and f preserves basepoints, then there is a relation between the homotopy groups of F, X, and Y in the form of a *long exact sequence*. We derive this by way of the *fiber sequence* associated to such an f.

Definition 0.1.1. A **pointed map** between pointed types (X, x_0) and (Y, y_0) is a map $f : X \to Y$ together with a path $f_0 : f(x_0) = y_0$.

For any pointed types (X, x_0) and (Y, y_0), there is a pointed map $(\lambda x.\, y_0) : X \to Y$ which is constant at the basepoint. We call this the **zero map** and sometimes write it as $0 : X \to Y$.

Recall that every pointed type (X, x_0) has a loop space $\Omega(X, x_0)$. We now note that this operation is functorial on pointed maps.

Definition 0.1.2. Given a pointed map between pointed types $f : X \to Y$, we define a pointed map $\Omega f : \Omega X \to \Omega Y$ by

$$(\Omega f)(p) :\equiv {f_0}^{-1} \cdot f(p) \cdot f_0.$$

The path $(\Omega f)_0 : (\Omega f)(\mathsf{refl}_{x_0}) = \mathsf{refl}_{y_0}$, which exhibits Ωf as a pointed map, is the obvious path of type

$${f_0}^{-1} \cdot f(\mathsf{refl}_{x_0}) \cdot f_0 = \mathsf{refl}_{y_0}.$$

There is another functor on pointed maps, which takes $f : X \to Y$ to $\mathsf{pr}_1 : \mathsf{fib}_f(y_0) \to X$. When f is pointed, we always consider $\mathsf{fib}_f(y_0)$ to be pointed with basepoint (x_0, f_0), in which case pr_1 is also a pointed map, with witness $(\mathsf{pr}_1)_0 :\equiv \mathsf{refl}_{x_0}$. Thus, this operation can be iterated.

Definition 0.1.3. The **fiber sequence** of a pointed map $f : X \to Y$ is the infinite sequence of pointed types and pointed maps

$$\ldots \xrightarrow{f^{(n+1)}} X^{(n+1)} \xrightarrow{f^{(n)}} X^{(n)} \xrightarrow{f^{(n-1)}} \ldots \longrightarrow X^{(2)} \xrightarrow{f^{(1)}} X^{(1)} \xrightarrow{f^{(0)}} X^{(0)}$$

defined recursively by

$$X^{(0)} :\equiv Y \qquad X^{(1)} :\equiv X \qquad f^{(0)} :\equiv f$$

and

$$\begin{aligned} X^{(n+1)} &:\equiv \mathsf{fib}_{f^{(n-1)}}(x_0^{(n-1)}) \\ f^{(n)} &:\equiv \mathsf{pr}_1 \qquad\qquad : X^{(n+1)} \to X^{(n)}. \end{aligned}$$

where $x_0^{(n)}$ denotes the basepoint of $X^{(n)}$, chosen recursively as above.

Thus, any adjacent pair of maps in this fiber sequence is of the form

$$X^{(n+1)} \equiv \mathsf{fib}_{f^{(n-1)}}(x_0^{(n-1)}) \xrightarrow{f^{(n)} \equiv \mathsf{pr}_1} X^{(n)} \xrightarrow{f^{(n-1)}} X^{(n-1)}.$$

In particular, we have $f^{(n-1)} \circ f^{(n)} = 0$. We now observe that the types occurring in this sequence are the iterated loop spaces of the base space Y, the total space X, and the fiber $F :\equiv \mathsf{fib}_f(y_0)$, and similarly for the maps.

Lemma 0.1.4. *Let $f : X \to Y$ be a pointed map of pointed spaces. Then:*

(i) The fiber of $f^{(1)} :\equiv \mathsf{pr}_1 : \mathsf{fib}_f(y_0) \to X$ is equivalent to ΩY.
(ii) Similarly, the fiber of $f^{(2)} : \Omega Y \to \mathsf{fib}_f(y_0)$ is equivalent to ΩX.
(iii) Under these equivalences, the pointed map $f^{(3)} : \Omega X \to \Omega Y$ is identified with the pointed map $\Omega f \circ (-)^{-1}$.

Proof. For (i), we have

$$\begin{aligned} \mathsf{fib}_{f^{(1)}}(x_0) &:\equiv \sum_{z:\mathsf{fib}_f(y_0)} (\mathsf{pr}_1(z) = x_0) \\ &\simeq \sum_{(x:X)} \sum_{(p:f(x)=y_0)} (x = x_0) && \text{(by original exercise 2.10)} \\ &\simeq (f(x_0) = y_0) && \text{(as } \textstyle\sum_{(x:X)}(x = x_0) \text{ is contractible)} \\ &\simeq (y_0 = y_0) && \text{(by } (f_0 \cdot -)\text{)} \\ &\equiv \Omega Y. \end{aligned}$$

Tracing through, we see that this equivalence sends $((x,p),q)$ to ${f_0}^{-1} \cdot f\big(q^{-1}\big) \cdot p$, while its inverse sends $r : y_0 = y_0$ to $((x_0, f_0 \cdot r), \mathsf{refl}_{x_0})$. In particular, the basepoint $((x_0,f_0),\mathsf{refl}_{x_0})$ of $\mathsf{fib}_{f^{(1)}}(x_0)$ is sent to ${f_0}^{-1} \cdot f\big({\mathsf{refl}_{x_0}}^{-1}\big) \cdot f_0$, which equals refl_{y_0}. Hence this equivalence is a pointed map (see original exercise 8.7). Moreover, under this equivalence, $f^{(2)}$ is identified with $\lambda r.\,(x_0, f_0 \cdot r) : \Omega Y \to \mathsf{fib}_f(y_0)$.

item (ii) follows immediately by applying (i) to $f^{(1)}$ in place of f. Since $(f^{(1)})_0 :\equiv \mathsf{refl}_{x_0}$, under this equivalence $f^{(3)}$ is identified with the map $\Omega X \to \mathsf{fib}_{f^{(1)}}(x_0)$ defined by $s \mapsto ((x_0,f_0),s)$. Thus, when we compose with the previous equivalence $\mathsf{fib}_{f^{(1)}}(x_0) \simeq \Omega Y$, we see that s maps to ${f_0}^{-1} \cdot f\big(s^{-1}\big) \cdot f_0$, which is by definition $(\Omega f)(s^{-1})$. We omit the proof that this is an equality of pointed maps rather than just of functions. □

Thus, the fiber sequence of $f : X \to Y$ can be pictured as:

$$\ldots \longrightarrow \Omega^2 X \xrightarrow{\Omega^2 f} \Omega^2 Y \xrightarrow{-\Omega\partial} \Omega F \xrightarrow{-\Omega i} \Omega X \xrightarrow{-\Omega f} \Omega Y \xrightarrow{\partial} F \xrightarrow{i} X \xrightarrow{f} Y.$$

where the minus signs denote composition with path inversion $(-)^{-1}$. Note that by original exercise 8.6, we have

$$\Omega\left(\Omega f \circ (-)^{-1}\right) \circ (-)^{-1} = \Omega^2 f \circ (-)^{-1} \circ (-)^{-1} = \Omega^2 f.$$

Thus, there are minus signs on the k-fold loop maps whenever k is odd.

From this fiber sequence we will deduce an *exact sequence of pointed sets*. Let A and B be sets and $f : A \to B$ a function, and recall from original definition 7.6.3 the definition of the *image* $\mathsf{im}(f)$, which can be regarded as a subset of B:

$$\mathsf{im}(f) :\equiv \{\, b : B \mid \exists(a : A).\, f(a) = b \,\}.$$

If A and B are moreover pointed with basepoints a_0 and b_0, and f is a pointed map, we define the **kernel** of f to be the following subset of A:

$$\ker(f) :\equiv \{\, x : A \mid f(x) = b_0 \,\}.$$

Of course, this is just the fiber of f over the basepoint b_0; it is a subset of A because B is a set.

Note that any group is a pointed set, with its unit element as basepoint, and any group homomorphism is a pointed map. In this case, the kernel and image agree with the usual notions from group theory.

Definition 0.1.5. An **exact sequence of pointed sets** is a (possibly bounded) sequence of pointed sets and pointed maps:

$$\ldots \longrightarrow A^{(n+1)} \xrightarrow{f^{(n)}} A^{(n)} \xrightarrow{f^{(n-1)}} A^{(n-1)} \longrightarrow \ldots$$

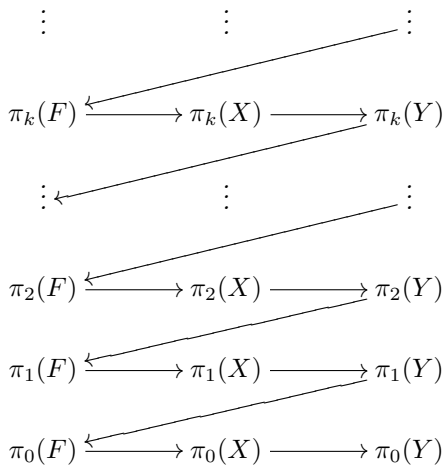
such that for every n, the image of $f^{(n)}$ is equal, as a subset of $A^{(n)}$, to the kernel of $f^{(n-1)}$. In other words, for all $a : A^{(n)}$ we have

$$(f^{(n-1)}(a) = a_0^{(n-1)}) \iff \exists(b : A^{(n+1)}).\, (f^{(n)}(b) = a).$$

where $a_0^{(n)}$ denotes the basepoint of $A^{(n)}$.

Usually, most or all of the pointed sets in an exact sequence are groups, and often abelian groups. When we speak of an **exact sequence of groups**, it is assumed moreover that the maps are group homomorphisms and not just pointed maps.

Theorem 0.1.6. *Let $f : X \to Y$ be a pointed map between pointed spaces with fiber $F :\equiv \mathsf{fib}_f(y_0)$. Then we have the following long exact sequence, which consists of groups except for the last three terms, and abelian groups except for the last six.*



Proof. We begin by showing that the 0-truncation of a fiber sequence is an exact sequence of pointed sets. Thus, we need to show that for any adjacent pair of maps in a fiber sequence:

$$\mathsf{fib}_f(z_0) \xrightarrow{g} W \xrightarrow{f} Z$$

with $g :\equiv \mathsf{pr}_1$, the sequence

$$\|\mathsf{fib}_f(z_0)\|_0 \xrightarrow{\|g\|_0} \|W\|_0 \xrightarrow{\|f\|_0} \|Z\|_0$$

is exact, i.e. that $\mathsf{im}(\|g\|_0) \subseteq \ker(\|f\|_0)$ and $\ker(\|f\|_0) \subseteq \mathsf{im}(\|g\|_0)$.

The first inclusion is equivalent to $\|f\|_0 \circ \|g\|_0 = 0$, which holds by functoriality of $\|-\|_0$ and the fact that $f \circ g = 0$. For the second, we assume $w' : \|W\|_0$ and $p' : \|f\|_0(w') = |z_0|_0$ and show there merely exists $t : \mathsf{fib}_f(z_0)$ such that $|g(t)|_0 = w'$. Since our goal is a mere proposition, we can assume that w' is of the form $|w|_0$ for some $w : W$. Now by original theorem 7.3.12, $p' : |f(w)|_0 = |z_0|_0$ yields $p'' : \|f(w) = z_0\|_{-1}$, so by a further truncation induction we may assume some $p : f(w) = z_0$. But now we have $|(w, p)|_0 : \|\mathsf{fib}_f(z_0)\|_0$ whose image under $\|g\|_0$ is $|w|_0 \equiv w'$, as desired.

Thus, applying $\|-\|_0$ to the fiber sequence of f, we obtain a long exact sequence involving the pointed sets $\pi_k(F)$, $\pi_k(X)$, and $\pi_k(Y)$ in the desired order. And of course, π_k is a group for $k \geq 1$, being the 0-truncation of a loop space, and an abelian group for $k \geq 2$ by the Eckmann–Hilton argument (original theorem 2.1.6). Moreover, Lemma 0.1.4 allows us to identify the maps $\pi_k(F) \to \pi_k(X)$ and $\pi_k(X) \to \pi_k(Y)$ in this exact sequence as $(-1)^k\pi_k(i)$ and $(-1)^k\pi_k(f)$ respectively.

More generally, every map in this long exact sequence except the last three is of the form $\|\Omega h\|_0$ or $\|-\Omega h\|_0$ for some h. In the former case it is a group homomorphism, while in the latter case it is a homomorphism if the groups are abelian; otherwise it is an "anti-homomorphism". However, the kernel and image of a group homomorphism are unchanged when we replace it by its negative, and hence so is the exactness of any sequence involving it. Thus, we can modify our long exact sequence to obtain one involving $\pi_k(i)$ and $\pi_k(f)$ directly and in which all the maps are group homomorphisms (except the last three). □